Probability

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ANSWER KEY — COMPLETE WORKED SOLUTIONS

Part I — Free-Response Questions (FRQs): Foundational Fluency (1–7)

1 **[Foundational]** • **Target: Basic probability**

Two fair six-sided dice are rolled. Find the probability their sum is 8.

Total outcomes: $n(S) = 36$. Favorable outcomes (sum $= 8$): $(2, 6), (3, 5), (4, 4), (5, 3), (6, 2)$ — that's 5.

2 **[Foundational]** • **Target: Permutations and combinations**

How many ways can a committee of 4 be chosen from 10 people?

Order doesn't matter, so use combinations: $\binom{10}{4} = \frac{10!}{4!\,6!} = \frac{10 \cdot 9 \cdot 8 \cdot 7}{4 \cdot 3 \cdot 2 \cdot 1} = 210$.

3 **[Foundational]** • **Target: Permutations and combinations**

How many ways can 2 officers (president, vice president — order matters) be chosen from 6 candidates?

Order matters, so use permutations: $P(6, 2) = \frac{6!}{4!} = 6 \cdot 5 = 30$.

4 **[Foundational]** • **Target: Permutations and combinations**

Explain, using Theorem 6.1, how $\binom{6}{2}$ relates to $P(6, 2)$ from the previous question.

Theorem 6.1 says $\binom{6}{2} = \frac{P(6,2)}{2!} = \frac{30}{2} = 15$ — combinations undercount permutations by exactly the $2!$ orderings of each chosen pair.

5 **[Foundational]** • **Target: Independent events and the addition rule**

Events A and B are independent, with $P(A) = 0.4$ and $P(B) = 0.5$. Find $P(A \text{ and } B)$.

By Definition 6.4, $P(A \text{ and } B) = P(A) \cdot P(B) = (0.4)(0.5) = 0.2$.

6 **[Foundational]** • **Target: Complement rule**

Events A and B satisfy $P(A) = 0.60$, $P(B) = 0.50$, and conditional probability $P(A|B) = 0.40$. Determine $P(A \cup B)$ and state whether A and B are independent.

Step 1: Compute intersection $P(A \cap B)$: Using the conditional probability multiplication rule: $P(A \cap B) = P(A|B) \cdot P(B) = 0.40 \times 0.50 = 0.20$. Step 2: Apply the general addition rule: $P(A \cup B) = P(A) + P(B) - P(A \cap B) = 0.60 + 0.50 - 0.20 = \mathbf{0.90}$. Step 3: Test for independence: $P(A|B) = 0.40$ while $P(A) = 0.60$. Since $P(A|B) eq P(A)$, events A and B are not independent.

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7 **[Foundational]** • **Target: Permutations and combinations**

A quality control inspector tests components that each have an independent 10% defect rate ($p = 0.10$). If a random batch of 5 components is tested: (a) Find the probability that exactly 2 components are defective. (b) Find the probability that at least 1 component is defective.

(a) Exactly 2 defective components: Apply the binomial probability formula $P(X = k) = \binom{n}{k} p^k (1-p)^{n-k}$ with $n = 5$, $k = 2$, $p = 0.10$: $P(X = 2) = \binom{5}{2}(0.10)^2(0.90)^3 = 10 \times 0.01 \times 0.729 = \mathbf{0.0729\ (7.29\%)}$. (b) At least 1 defective component: Use complementary probability: $P(X \geq 1) = 1 - P(X = 0) = 1 - \binom{5}{0}(0.10)^0(0.90)^5 = 1 - (1)(1)(0.59049) = 1 - 0.59049 = \mathbf{0.40951\ (40.95\%)}$.

Part II — Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

8 **[Intermediate]** • **Target: Independent events and the addition rule**

Two events A and B are independent. $P(A) = 0.4$, $P(B) = 0.5$. Find $P(A \cap B)$.

Step 1: State definition of statistical independence: Two events are independent if and only if: $P(A \cap B) = P(A) \cdot P(B)$. Step 2: Compute joint probability: $P(A \cap B) = 0.4 \times 0.5 = \mathbf{0.20}$.

9 **[Intermediate]** • **Target: Basic probability**

A delegation of 5 delegates is chosen from 6 men and 8 women. (a) In how many ways can the delegation be selected if it must contain exactly 3 women? (b) What is the exact probability that the delegation contains at least 4 women?

(a) Exactly 3 women: Choose 3 women from 8 and 2 men from 6: $\binom{8}{3} \times \binom{6}{2} = \frac{8 \times 7 \times 6}{6} \times \frac{6 \times 5}{2} = 56 \times 15 = \mathbf{840}$ ways. (b) Probability of at least 4 women:Total possible 5-delegate committees from all 14 people: $\binom{14}{5} = \frac{14 \times 13 \times 12 \times 11 \times 10}{5 \times 4 \times 3 \times 2 \times 1} = 2002$. Case 1 (4 women, 1 man): $\binom{8}{4}\binom{6}{1} = 70 \times 6 = 420$.Case 2 (5 women, 0 men).

10 **[Intermediate]** • **Target: Complement rule**

Find the constant term (independent of x) in the binomial expansion of $\left(2x^2 - \frac{1}{x}\right)^9$.

Step 1: Write the general term T_{k+1}: $T_{k+1} = \binom{9}{k}(2x^2)^{9-k}\left(-\frac{1}{x}\right)^k = \binom{9}{k}2^{9-k}x^{2(9-k)}(-1)^k x^{-k} = \binom{9}{k}2^{9-k}(-1)^k x^{18-3k}$. Step 2: Find k for the term independent of x: Set exponent equal to 0: $18 - 3k = 0 \Rightarrow 3k = 18 \Rightarrow k = 6$. Step 3: Evaluate coefficient at $k = 6$: $\binom{9}{6} = \binom{9}{3} = \frac{9 \times 8 \times 7}{3 \times 2 \times 1} = 84$. .

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11 **[Intermediate]** • **Target: Permutations and combinations**

A dataset of 40 exam scores has mean $\bar{x} = 72$ and sample standard deviation $s = 8$. The instructor rescales all scores using $y_i = 1.25x_i - 10$. What are the mean and standard deviation of the rescaled scores?

Step 1: Compute transformed mean: For a linear transformation $y = ax + b$, the mean transforms identically: $\bar{y} = 1.25\bar{x} - 10 = 1.25(72) - 10 = 90 - 10 = \mathbf{80}$. Step 2: Compute transformed standard deviation: The additive constant -10 shifts the entire distribution without changing spread. Only the multiplicative factor scales dispersion: $s_y = |a| \cdot s_x = 1.25 \times 8 = \mathbf{10}$.

12 **[Intermediate]** • **Target: Independent events and the addition rule**

Find all complex solutions to $z^4 + 16 = 0$ in exact rectangular form $a + bi$ using De Moivre's roots theorem.

Step 1: Express -16 in polar modulus-argument form: $z^4 = -16 = 16(\cos\pi + i\sin\pi)$. Here $r = 16$, $\theta = \pi$, and $n = 4$. Step 2: Apply n-th roots formula: $w_k = \sqrt[4]{16}\left[\cos\left(\frac{\pi+2\pi k}{4}\right) + i\sin\left(\frac{\pi+2\pi k}{4}\right)\right] = 2\left[\cos\left(\frac{(2k+1)\pi}{4}\right) + i\sin\left(\frac{(2k+1)\pi}{4}\right)\right],\ k \in \{0,1,2,3\}$. Step 3: Evaluate for each k:$k = 0$.

13 **[Intermediate]** • **Target: Basic probability**

Find all distinct points of intersection between the two polar curves $r = 2\sin\theta$ and $r = 2\cos\theta$ on $0 \le \theta < 2\pi$.

Equate the two equations: $2\sin\theta = 2\cos\theta \Rightarrow \tan\theta = 1$. On $[0, 2\pi)$, $\theta = \frac{\pi}{4}$ or $\theta = \frac{5\pi}{4}$.For $\theta = \frac{\pi}{4}$: $r = 2\sin\left(\frac{\pi}{4}\right) = \sqrt{2}$, giving point $\left(\sqrt{2}, \frac{\pi}{4}\right)$.For $\theta = \frac{5\pi}{4}$: $r = 2\sin\left(\frac{5\pi}{4}\right) = -\sqrt{2}$. However, $\left(-\sqrt{2}, \frac{5\pi}{4}\right)$ represents the identical geometric point as $\left(\sqrt{2}, \frac{\pi}{4}\right)$.Next, check the pole.

14 **[Intermediate]** • **Target: Complement rule**

Compute $(\sqrt{3} - i)^8$ using De Moivre's Theorem, expressing the result in standard rectangular form $a + bi$.

Step 1: Polar form of $z = \sqrt{3} - i$: Modulus: $r = \sqrt{(\sqrt{3})^2 + (-1)^2} = \sqrt{3+1} = 2$. Argument: In Quadrant IV with $\tan\theta = -\frac{1}{\sqrt{3}} \Rightarrow \theta = -\frac{\pi}{6}$. $z = 2\left(\cos\left(-\frac{\pi}{6}\right) + i\sin\left(-\frac{\pi}{6}\right)\right)$. Step 2: Apply De Moivre's Theorem.

Part III — Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

15 **[Challenging]** • **Target: Permutations and combinations** • **Cross-Lesson: 5.01 Introductory Calculus**

Decompose the vector $\mathbf{u} = \langle 6, 2\rangle$ into components parallel and orthogonal to $\mathbf{v} = \langle 3, 4\rangle$.

Step 1: Compute dot product and squared magnitude: $\mathbf{u} \cdot \mathbf{v} = (6)(3) + (2)(4) = 18 + 8 = 26$. $||\mathbf{v}||^2 = 3^2 + 4^2 = 9 + 16 = 25$. Step 2: Vector projection (parallel component): $\mathbf{u}_{\parallel} = \text{proj}_{\mathbf{v}}\mathbf{u} = \left(\frac{\mathbf{u}\cdot\mathbf{v}}{||\mathbf{v}||^2}\right)\mathbf{v} = \frac{26}{25}\langle 3, 4\rangle = \left\langle \frac{78}{25}, \frac{104}{25}\right\rangle = \langle 3.12, 4.16\rangle$. Step 3: Orthogonal component.

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16 **[Challenging]** • **Target: Independent events and the addition rule** • **Cross-Lesson: 1.01 Complex Numbers**

Consider the points $A(1, 2, 3)$, $B(2, 4, 1)$, and $C(-1, 0, 5)$ in $\mathbb{R}^3$. Find a non-zero vector $\mathbf{n}$ normal to the plane containing A, B, C, and write the Cartesian equation of the plane.

Step 1: Form vectors $\vec{AB}$ and $\vec{AC}$: $\vec{AB} = \langle 2-1, 4-2, 1-3 \rangle = \langle 1, 2, -2 \rangle$.
$\vec{AC} = \langle -1-1, 0-2, 5-3 \rangle = \langle -2, -2, 2 \rangle$. Step 2: Compute normal vector via cross product:
$\mathbf{n} = \vec{AB} \times \vec{AC} = |\mathbf{i}, \mathbf{j}, \mathbf{k}; 1, 2, -2; -2, -2, 2| = \mathbf{i}(4-4) - \mathbf{j}(2-4) + \mathbf{k}(-2-(-4)) = 0\mathbf{i} + 2\mathbf{j} + 2\mathbf{k} = \langle 0, 2, 2 \rangle$. Simplifying by scalar.

17 **[Challenging]** • **Target: Basic probability** • **Cross-Lesson: 2.01 Sequences & Series**

Prove by mathematical induction that for all integers $n \geq 1$: $1 + 3 + 5 + \ldots + (2n-1) = n^2$.

1. Base Step ($n = 1$): LHS $= 2(1) - 1 = 1$, RHS $= 1^2 = 1$. Since LHS = RHS, the base case $P(1)$ is true. 2. Inductive Hypothesis: Assume $P(k)$ is true for some integer $k \geq 1$:
$1 + 3 + 5 + \ldots + (2k-1) = k^2$. 3. Inductive Step ($n = k+1$): Show that
$1 + \ldots + (2k-1) + (2(k+1)-1) = (k+1)^2$: LHS $= [1 + 3 + \ldots + (2k-1)] + (2k+1)$. By the inductive hypothesis, substitute k^2.

18 **[Challenging]** • **Target: Complement rule** • **Cross-Lesson: 3.01 Functions & Graphs**

Solve the trigonometric equation $\cos(2\theta) + 3\sin\theta = 2$ for all solutions $\theta \in [0, 2\pi)$.

Substitute $\cos(2\theta) = 1 - 2\sin^2\theta$ to express the entire equation in terms of $\sin\theta$:
$(1 - 2\sin^2\theta) + 3\sin\theta = 2$. $-2\sin^2\theta + 3\sin\theta - 1 = 0 \Rightarrow 2\sin^2\theta - 3\sin\theta + 1 = 0$. Factor the quadratic in $\sin\theta$:
$(2\sin\theta - 1)(\sin\theta - 1) = 0$. Case 1: $2\sin\theta - 1 = 0 \Rightarrow \sin\theta = \frac{1}{2}$. On $[0, 2\pi)$, $\theta = \frac{\pi}{6}$ or $\theta = \frac{5\pi}{6}$. Case 2:
$\sin\theta - 1 = 0 \Rightarrow \sin\theta = 1$. On.

19 **[Challenging]** • **Target: Permutations and combinations** • **Cross-Lesson: 4.01 Matrices & Vectors**

In triangle ABC, $a = 10$, $b = 16$, and $A = 30^\circ$. Determine whether $0, 1,$ or 2 distinct triangles can be formed, and calculate all possible values for angle B to the nearest tenth of a degree.

Step 1: Set up the Law of Sines for angle B: $\frac{\sin B}{b} = \frac{\sin A}{a} \Rightarrow \sin B = \frac{b \sin A}{a} = \frac{16 \sin 30^\circ}{10} = \frac{16(0.5)}{10} = \frac{8}{10} = 0.8$. Step 2: Find the acute angle solution B_1: $B_1 = \sin^{-1}(0.8) \approx \mathbf{53.1}^\circ$. Step 3: Test the obtuse supplementary angle B_2:
$B_2 = 180^\circ - 53.1^\circ = 126.9^\circ$. Check triangle sum with given $A = 30^\circ$: \[A + B_2 = 30^\circ + 126.9^\circ = 156.9^\circ Conclusion: Exactly **2** d.

20 **[Challenging]** • **Target: Independent events and the addition rule** • **Cross-Lesson: 5.01 Introductory Calculus**

In triangle ABC, $a = 7$, $b = 5$, and $\cos C = \frac{1}{5}$. (a) Find the exact length of side c. (b) Calculate the exact value of $\sin^2 C$ and the area of triangle ABC.

Part (a): Find side c using Law of Cosines: $c^2 = a^2 + b^2 - 2ab\cos C$
$c^2 = 7^2 + 5^2 - 2(7)(5)\left(\frac{1}{5}\right) = 49 + 25 - 14 = 60 \Rightarrow \mathbf{c = \sqrt{60} = 2\sqrt{15}}$. Part (b): Find $\sin^2 C$ and Area: 1. Pythagorean identity:
$\sin^2 C = 1 - \cos^2 C = 1 - \left(\frac{1}{5}\right)^2 = 1 - \frac{1}{25} = \mathbf{\frac{24}{25}}$. 2. Since C is an angle of a triangle,
$\sin C > 0 \Rightarrow \sin C = \frac{\sqrt{24}}{5} = \frac{2\sqrt{6}}{5}$. 3. Area formula.

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Part IV — Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

21 [Foundational] • Target: Basic probability

A bag contains 9 yellow marbles, 5 red marbles, and 7 blue marbles. A marble is drawn at random. Find $P(\text{yellow})$.

a. $\frac{1}{3}$ **b.** $\frac{3}{7}$ ✓

c. $\frac{10}{21}$ **d.** $\frac{5}{21}$

Total marbles: $n(S) = 21$. $P(\text{yellow}) = \frac{9}{21} = \frac{3}{7}$.

22 [Foundational] • Target: Complement rule

A bag contains 3 green marbles, 7 yellow marbles, and 7 red marbles. A marble is drawn at random. Find $P(\text{green})$.

a. $\frac{7}{17}$ **b.** $\frac{5}{17}$

c. $\frac{4}{17}$ **d.** $\frac{3}{17}$ ✓

Total marbles: $n(S) = 17$. $P(\text{green}) = \frac{3}{17}$.

23 [Foundational] • Target: Permutations and combinations

A bag contains 7 blue marbles, 7 green marbles, and 3 yellow marbles. A marble is drawn at random. Find $P(\text{blue})$.

a. $\frac{9}{17}$ **b.** $\frac{8}{17}$

c. $\frac{3}{17}$ **d.** $\frac{7}{17}$ ✓

Total marbles: $n(S) = 17$. $P(\text{blue}) = \frac{7}{17}$.

24 [Foundational] • Target: Independent events and the addition rule

A bag contains 3 red marbles, 3 blue marbles, and 4 green marbles. A marble is drawn at random. Find $P(\text{red})$.

a. $\frac{3}{10}$ ✓ **b.** $\frac{1}{2}$

c. $\frac{3}{5}$ **d.** $\frac{2}{5}$

Total marbles: $n(S) = 10$. $P(\text{red}) = \frac{3}{10}$.

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25 [Foundational] • Target: Basic probability

How many ways can 2 distinct positions be filled from 12 candidates, where order matters?

a. 66

b. 138

c. 132 ✓

d. 126

Order matters, so use permutations: $P(12, 2) = \frac{12!}{10!} = 12 \times 11 = 132$.

26 [Foundational] • Target: Complement rule

Out of 38 equally likely tiles, 5 are red, 8 are numbered, and 3 are both. One item is chosen at random. Find $P(\text{red or numbered})$.

a. $\frac{5}{19}$ ✓

b. $\frac{6}{19}$

c. $\frac{13}{38}$

d. $\frac{4}{19}$

By the addition rule: $P(\text{red or numbered}) = \frac{5}{38} + \frac{8}{38} - \frac{3}{38} = \frac{10}{38} = \frac{5}{19}$.

27 [Foundational] • Target: Permutations and combinations

Out of 27 equally likely boxes, 7 are labeled, 8 are sealed, and 7 are both. One item is chosen at random. Find $P(\text{labeled or sealed})$.

a. $\frac{2}{9}$

b. $\frac{10}{27}$

c. $\frac{8}{27}$ ✓

d. $\frac{5}{9}$

By the addition rule: $P(\text{labeled or sealed}) = \frac{7}{27} + \frac{8}{27} - \frac{7}{27} = \frac{8}{27}$.

28 [Foundational] • Target: Independent events and the addition rule

Out of 26 equally likely tickets, 9 are gold, 12 are striped, and 8 are both. One item is chosen at random. Find $P(\text{gold or striped})$.

a. $\frac{21}{26}$

b. $\frac{15}{26}$

c. $\frac{1}{2}$ ✓

d. $\frac{11}{26}$

By the addition rule: $P(\text{gold or striped}) = \frac{9}{26} + \frac{12}{26} - \frac{8}{26} = \frac{13}{26} = \frac{1}{2}$.

Part V — Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

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29 [Intermediate] • Target: Basic probability

Out of 37 equally likely numbers, 4 are even, 10 are prime, and 2 are both. One item is chosen at random. Find $P(\text{even or prime})$.

a. $\frac{14}{37}$

b. $\frac{10}{37}$

c. $\frac{12}{37}$ ✓

d. $\frac{15}{37}$

By the addition rule: $P(\text{even or prime}) = \frac{4}{37} + \frac{10}{37} - \frac{2}{37} = \frac{12}{37}$.

30 [Intermediate] • Target: Complement rule

Events A and B are mutually exclusive, with $P(A) = 2/7$ and $P(B) = 3/8$. Find $P(A \text{ or } B)$.

a. **37/56** ✓

b. 269/280

c. 55/168

d. 83/168

Mutually exclusive events: $P(A \text{ or } B) = P(A) + P(B) = 2/7 + 3/8 = 37/56$.

31 [Intermediate] • Target: Permutations and combinations

A survey was conducted on the preferred modes of transportation for students travelling to school. The results are summarized in the two-way table below. If a student is chosen at random from those surveyed, and it is known that they prefer to travel by bus, what is the probability that they are in Year 10? Let N_{10} represent the number of Year 10 students who prefer the bus, and N_{Bus} represent the total number of students who prefer the bus. The probability we are looking for is $\frac{N_{10}}{N_{Bus}}$. | Year Group | Bicycle | Bus | Train | Walk | Total | |---|---|---|---|---|---| | Year 9 | 45 | 60 | 25 | 70 | 200 | | Year 10 | 55 | x | 35 | 80 | 270 | | Year 11 | 30 | 50 | 40 | 50 | 170 | | Total | 130 | 160 | 100 | 200 | 640 |

a. $\frac{130}{160}$

b. $\frac{x}{160}$ ✓

c. $\frac{x}{270}$

d. $\frac{160}{640}$

The question asks for the conditional probability that a student is in Year 10, given that they prefer to travel by bus. This can be represented as $P(\text{Year 10}|\text{Bus})$. Using the formula for conditional probability, $P(A|B) = \frac{P(A \cap B)}{P(B)}$. In this context, A is the event that a student is in Year 10, and B is the event that a student prefers to travel by bus. The number of students in Year 10 who prefer the bus is given as x in the table. The total number of students who prefer.

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32 [Intermediate] • Target: Independent events and the addition rule

In triangle ABC, point D lies on side BC such that AD is an angle bisector of angle A. Let $AB = c$, $AC = b$, $BD = m$, and $CD = n$. Prove that $\frac{m}{n} = \frac{c}{b}$ using a combination of the Sine Rule and the properties of angle bisectors.

a. This statement is a direct consequence of the Sine Rule applied to triangles ABD and ACD. ✓

b. This statement can be proven by applying the Law of Cosines to triangles ABD and ACD and equating expressions for AD^2.

c. This statement is true only if triangle ABC is isosceles with $b = c$.

d. This statement requires the use of vector methods for its proof.

Let $\angle BAD = \angle CAD = \alpha$. In triangle ABD, by the Sine Rule, we have $\frac{m}{\sin \alpha} = \frac{c}{\sin \angle ADB}$. Therefore, $m = \frac{c \sin \alpha}{\sin \angle ADB}$. In triangle ACD, by the Sine Rule, we have $\frac{n}{\sin \alpha} = \frac{b}{\sin \angle ADC}$. Therefore, $n = \frac{b \sin \alpha}{\sin \angle ADC}$. Since $\angle ADB$ and $\angle ADC$ are supplementary angles, $\sin \angle ADB = \sin(180^\circ - \angle ADC) = \sin \angle ADC$. Now, we can form the ratio.

33 [Intermediate] • Target: Basic probability

The bar chart below shows the number of students who chose different sports as their favorite. What is the total number of students surveyed?

a. 50

b. 60

c. 70

d. 80 ✓

To find the total number of students surveyed, we need to sum the number of students for each sport. From the bar chart: - Soccer: 20 students - Basketball: 15 students - Tennis: 10 students - Swimming: 25 students Total students = 20 + 15 + 10 + 25 = 70 Therefore, the total number of students surveyed is 70.

34 [Intermediate] • Target: Complement rule

The following bar chart shows the number of students who prefer different fruits. What is the total number of students who prefer apples and bananas?

a. 20

b. 25 ✓

c. 30

d. 35

From the bar chart, the number of students who prefer apples is 15. The number of students who prefer bananas is 10. The total number of students who prefer apples and bananas is the sum of these two values: 15 + 10 = 25.

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35 [Intermediate] • Target: Permutations and combinations

The bar chart below shows the number of students who achieved different scores on a recent quiz. What is the total number of students who scored 8 or 10?

a. 25

b. **30 ✓**

c. 35

d. 40

The bar chart shows that 15 students scored 8 and 20 students scored 10. Therefore, the total number of students who scored 8 or 10 is $15 + 20 = 35$.

36 [Intermediate] • Target: Independent events and the addition rule

The table shows exam marks for 20 students. Find the mean and identify the modal class.

a. **Mean ≈ 65, Modal class 60–69 ✓**

b. Mean ≈ 68, Modal class 70–79

c. Mean ≈ 60, Modal class 50–59

d. Mean ≈ 72, Modal class 60–69

Mid-points: 44.5(2)+54.5(4)+64.5(7)+74.5(5)+84.5(2) = 89+218+451.5+372.5+169 = 1300; mean = 1300/20 = 65. Modal class = 60–69 (highest frequency).

37 [Intermediate] • Target: Basic probability

In a group of 20 students, 9 study French, and 1 study both French and Spanish. If a student is chosen at random from those studying French, what is the probability they also study Spanish?

a. **1/9 ✓**

b. 7/36

c. 19/90

d. 5/18

$P(\text{Spanish} \mid \text{French}) = \frac{P(\text{both})}{P(\text{French})} = \frac{1/20}{9/20} = \frac{1}{9} = 1/9$.

Part VI — Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

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38 [Challenging] • Target: Complement rule • Cross-Lesson: 3.01 Functions & Graphs

A bag has 23 marbles, of which 2 are red. What is the probability of drawing a red marble?

a. 43/230

b. **2/23** ✓

c. 89/230

d. 33/115

Applying Statistics & Probability principles, $P(\text{red}) = \frac{2}{23}$. Therefore, the verified correct option is 2/23.

39 [Challenging] • Target: Permutations and combinations • Cross-Lesson: 4.01 Matrices & Vectors

In how many ways can 2 of 7 books on a shelf be arranged?

a. 82

b. 33

c. **42** ✓

d. 341

Applying Statistics & Probability principles, $7P2 = \frac{7!}{(7-2)!} = 42$. Therefore, the verified correct option is 42.

40 [Challenging] • Target: Independent events and the addition rule • Cross-Lesson: 5.01 Introductory Calculus

In how many ways can 3 of 4 photos in an album be arranged?

a. 132

b. 69

c. 154

d. **24** ✓

Applying Statistics & Probability principles, $4P3 = \frac{4!}{(4-3)!} = 24$. Therefore, the verified correct option is 24.

41 [Challenging] • Target: Basic probability • Cross-Lesson: 1.01 Complex Numbers

In how many ways can 2 of 4 runners in a race (1st through r-th place) be arranged?

a. 296

b. 112

c. **12** ✓

d. 270

Applying Statistics & Probability principles, $4P2 = \frac{4!}{(4-2)!} = 12$. Therefore, the verified correct option is 12.

42 [Challenging] • Target: Complement rule • Cross-Lesson: 2.01 Sequences & Series

Find the population variance of the data set: $2, 8, 12, 8, 10$.

a. 173/15

b. **56/5** ✓

c. 677/60

d. 163/15

Mean $= 8$. Variance $= \frac{(2-8)^2, (8-8)^2, (12-8)^2, (8-8)^2, (10-8)^2}{5} = \frac{56}{5} = 56/5$.

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43 [Challenging] • Target: Permutations and combinations • Cross-Lesson: 3.01 Functions & Graphs

A basketball player makes a free throw with probability $2/3$. If she takes 2 free throws, what is the probability she makes at least one?

a. 53/90

b. 55/72

c. **8/9 ✓**

d. 35/36

$P(\text{at least one}) = 1 - P(\text{none}) = 1 - (1/3)^2 = 1 - 1/9 = 8/9$.

44 [Challenging] • Target: Independent events and the addition rule • Cross-Lesson: 4.01 Matrices & Vectors

A committee of 5 people is chosen from 9 people, and then 4 of those 5 are assigned distinct roles (e.g., chair, secretary, ...). How many ways can this be done?

a. 2696

b. **15120 ✓**

c. 396

d. 4484

Choose the committee: 95. Assign roles: $5P4$. Total $= 9\text{5}P4 = 126120 = 15120$.

45 [Challenging] • Target: Basic probability • Cross-Lesson: 5.01 Introductory Calculus

Find the median of the data set: $9, 4, 9, 2, 7$.

a. 17

b. 24

c. 13

d. **7 ✓**

Sorted: $2, 4, 7, 9, 9$. With 5 values, the median is the middle value: 7.

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